UK Industrial Energy Competitiveness

From Price Relief to Structural Resilience

A proposal for a long-term mechanism to secure the energy competitiveness of British industry

Prepared for discussion with the Department for Energy Security and Net Zero (DESNZ), industry associations and energy-market participants.

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Status: Analytical proposal for discussion. Not commissioned government advice.

Executive Summary

British industry faces a structural cost disadvantage in electricity. In H2 2025, large industrial consumers in the 20–70 GWh/year band paid 16.88 p/kWh before taxes and levies and 23.37 p/kWh including them.

On the European comparison, the UK pre-tax price was 42% above Germany, 71% above the Netherlands and 148% above France. Including taxes and levies, the gaps widened to 68%, 130% and 214%. A separate IEA benchmark for very large industrial consumers (>150 GWh/year) places selected 2024 regional examples in China, India and the United States at approximately US$70, US$62 and US$45/MWh. The global cohort and period differ, so these are indicative context, not like-for-like comparisons.

The economic transmission is direct: higher electricity costs raise unit production costs; higher costs compress margins or force higher selling prices; British producers then lose price competitiveness, orders and market share. Persistent loss of orders reduces investment, capacity utilisation, skills and supplier ecosystems. Producers combining lower-cost dependable energy with automation can undercut British output.

Germany is a useful European comparator, not a sufficient strategic target. German energy-intensive output has also been under pressure. Matching Germany would not by itself secure the future competitiveness of British industry.

Five measures — and the problem each addresses

Measure	Problem addressed	Proposed response
1. Long-term industrial power contracts	Short procurement horizons and wholesale exposure undermine cost visibility and investment.	Enable 10–20-year contracts with transparent price, delivery, balancing, indexation and risk allocation.
2. Firm low-carbon supply and resilience	Gas-linked price formation and insufficient dependable low-carbon capacity expose industry to shocks.	Support safe nuclear life-extension and disciplined new build, alongside storage, interconnection, reserves and demand response.
3. Renewables assessed on full system value	Standalone LCOE can omit balancing, networks, curtailment, land and end-of-life liabilities.	Assess delivered system value, firm-capacity contribution, land opportunity cost, lifecycle O&M and decommissioning.
4. International energy portfolio	Domestic industry remains exposed to external fuel and infrastructure shocks.	Assess diversified assets while distinguishing financial returns from enforceable physical supply rights.
5. Industrial zones and stabilisation fund	Temporary relief does not solve connection, infrastructure coordination or extreme price risks.	Coordinate investment around industrial clusters; provide narrowly defined, ring-fenced and stress-tested support.

A 24–36-month pilot should focus on data, contract design, governance, stress testing and limited site-level measures. It should not promise completion of major generation or transmission projects within that period.

Author, Scope and Status

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His work includes systems architecture, industrial software, data analytics, knowledge-management systems and security engineering. Alongside his engineering practice, he has accumulated experience in public and political processes since 2017.

This proposal applies a systems-engineering approach to industrial energy policy. The author does not present himself as an energy-market economist or specialist in nuclear or renewable generation. Findings based on public data require validation by relevant technical, economic and institutional experts. The author has no commercial interest in a particular generation technology, supplier or industrial sector.

Prepared for discussion with DESNZ, industry associations and energy-market participants. Analytical proposal for discussion; not commissioned government advice. Prepared with analytical assistance from ChatGPT and Grok.

1. The Problem and Its Economic Consequences

For energy-intensive industries, electricity is a determinant of unit cost and the ability to compete. Exposed sectors include metals, chemicals, fertilisers, cement, glass, paper and selected branches of engineering.

The transmission mechanism is cumulative: higher energy cost → higher production cost → lower margins or higher selling prices → loss of orders and market share → reduced investment and capacity utilisation → loss of skills, suppliers and industrial capability. Once specialist teams and supplier ecosystems disappear, restoring them can cost more than preserving them.

ONS reported that output of four energy-intensive manufacturing groups fell by 33.6% between 2021 Q1 and 2024 Q4, reaching the lowest point in the available series beginning in 1990. Energy is not the sole cause of industrial contraction; it is a material constraint, not a single-cause explanation.

Destatis reported that from February 2022 to March 2026, production in German energy-intensive branches fell by 15.2%, compared with a 9.5% decline in total industry. Employment in those branches fell by 6.3%, approximately 53,200 people. These figures do not establish that energy prices alone caused the decline.

2. UK Position Relative to Competitors

The table combines the European comparison and global benchmark. European values cover 20–70 GWh/year consumers in H2 2025. IEA global values cover selected regional examples for very large energy-intensive consumers, generally above 150 GWh/year, in 2024. Global percentages against the UK are arithmetic context only, not controlled like-for-like comparisons.

Country / region	US$/MWh	Basis / period	UK difference
United Kingdom	226.2 pre-tax; 313.2 incl. taxes	20–70 GWh/year, H2 2025	Reference
Germany	159.1 pre-tax; 186.1 incl. taxes	20–70 GWh/year, H2 2025	+42% pre-tax; +68% incl.
Netherlands	132.4 pre-tax; 136.1 incl. taxes	20–70 GWh/year, H2 2025	+71%; +130%
France	91.3 pre-tax; 99.6 incl. taxes	20–70 GWh/year, H2 2025	+148%; +214%
EU	106	IEA regional benchmark, 2024; >150 GWh/year	about +113%*
Italy	116	IEA regional benchmark, 2024; >150 GWh/year	about +95%*
Germany	91	IEA regional benchmark, 2024; >150 GWh/year	about +148%*
France	73	IEA regional benchmark, 2024; >150 GWh/year	about +210%*
Norway	28	IEA regional benchmark, 2024; >150 GWh/year	about +708%*
United States (Texas)	45	IEA regional example, 2024; >150 GWh/year	about +403%*
China (Inner Mongolia)	70	IEA regional example, 2024; >150 GWh/year	about +223%*
India (Andhra Pradesh)	62	IEA regional example, 2024; >150 GWh/year	about +265%*

* Indicative arithmetic against UK H2 2025 pre-tax reference US$226.2/MWh; cohorts, periods, geography and methodology differ.

European source values in p/kWh: UK 16.88 / 23.37; Germany 11.87 / 13.89; Netherlands 9.88 / 10.16; France 6.81 / 7.43. Conversion uses 2025 annual-average GBP/USD of approximately 1.34. Russia is not assigned a value because a defensible comparison requires a defined regional industrial end-user tariff, consumption profile, date and tax basis.

The global comparison does not claim that the average British factory pays the same price as a factory in Inner Mongolia, Andhra Pradesh or Texas. It illustrates the scale of the international cost challenge.

3. Structural Causes of the Price Gap

3.1 Gas exposure and wholesale price formation

Under marginal electricity pricing, gas-fired generation can set the wholesale price when required to meet demand. Gas prices can therefore influence electricity prices even when a significant share of annual generation comes from nuclear or renewables.

3.2 Firm capacity and system adequacy

Industrial users need power when production schedules require it. Wind and solar contribute annual energy but are variable and not fully dispatchable. Their firm-capacity contribution is distinct from nameplate capacity and annual capacity factor. Nuclear can provide dependable low-carbon generation, but planned maintenance, unplanned outages and construction delays require reserves and diversity.

3.3 Networks, balancing and curtailment

NESO spent £1.34bn on direct thermal constraint actions in 2024/25. The government's Reformed National Pricing Delivery Plan states that, without further mitigating action, constraint costs could peak around £7bn in 2030/31, depending on network build and generation deployment scenarios. These are scenario-dependent costs, not an inevitable outcome.

3.4 Policy costs and levies

IFS analysis for industrial consumers without an energy-intensive industry discount finds that between 2017 and 2025 (2025 prices), energy and supply costs increased by 3.1 p/kWh, network costs fell by 2.2 p/kWh, and taxes, fees and levies increased by 4.5 p/kWh. The net increase across these components is 5.4 p/kWh. Thus the levy increase represents about 83% of the net increase. This is a contribution-to-net-change ratio, not the percentage growth rate of levies themselves.

The IFS headline 59% real increase in taxes and levies applies to household bills, not this industrial group. Applying it to industrial consumers would be misleading. The cited decomposition does not provide a directly comparable industrial percentage growth rate.

3.5 Contract structure

Long-term power purchase agreements and hedging can reduce exposure to spot volatility, but cannot eliminate generation, network, balancing or counterparty costs. Contract design should allocate risks transparently.

3.6 Renewables: efficiency, land and system value

Module efficiency, capacity factor, annual energy yield and firm-capacity contribution are different measures. JRC reports average PV module efficiency increased from 16.3% in 2014 to 22.6% in 2024. This is material progress, but efficiency gains alone do not remove intermittency, seasonal variation, storage, transmission or land-use requirements.

EN-3 (2025) estimates utility-scale solar farms, including associated infrastructure, at approximately 1.6–2.25 hectares per MW, with site variation. Local competition for agricultural, industrial and recreational land must be assessed. For wind, the entire turbine-spacing envelope is not permanently unavailable to agriculture; direct footprint, roads and construction disturbance should be distinguished from the wider project area.

Project appraisal should account for connection, transmission, curtailment, balancing, storage, firm-capacity contribution, land use, lifecycle O&M and decommissioning. Rooftops, car parks, brownfield sites, industrial land, offshore generation and co-location may reduce some conflicts, but are site-specific.

4. Renewable Infrastructure: Full Lifecycle Cost and End-of-Life Liability

Appraisal should disclose asset life by component, planned and unplanned maintenance, major component replacement, repowering or life extension, decommissioning, transport, recycling or disposal, site restoration and the financial security available to meet those obligations.

Asset / component	Indicative life	Key lifecycle issue
Solar PV modules	25–35 years	Degradation, inspection, replacement and end-of-life processing.
String inverters	10–15 years	One or more replacements may be needed during panel life.
Microinverters	20–25 years	Different replacement profile from string inverter systems.
Lithium-ion batteries	10–15 years	Capacity fade, augmentation, replacement and recycling.
Onshore wind turbines	About 25–32 years; possible extensions	Condition-based maintenance and major component replacement.
Offshore wind farms	About 25–30 years	Higher vessel, access, weather-window, insurance and maintenance costs.

IRENA’s global weighted-average utility-scale PV O&M benchmark is about US$13.1/kW/year. At 1 GW this is about US$13.1m annually, or US$328m over 25 years undiscounted at constant cost. It is not a UK project quote and does not include every replacement, financing or grid cost.

DESNZ/Arup’s 2025 offshore wind assessment gives median O&M of about £46,500/MW/year (2023 prices), range £30,500–£64,600. For 1 GW, the median is £46.5m/year or £1.16bn over 25 years undiscounted. Broader OPEX, including insurance and grid charges, is higher and must not be labelled maintenance alone.

A DESNZ peer review recommends offshore wind decommissioning planning scenarios of approximately £110,000, £255,000 and £507,000 per MW. For 1 GW, these imply £110m, £255m and £507m. Scope, foundations, cables, vessels, seabed obligations and scrap recovery drive variation. Onshore UK evidence is less robust; blade management can be a significant unresolved cost.

Wind blades are composite structures that are harder to recycle than metals. Reported UK recycling gate fees range approximately £200–£1,500 per tonne. Reuse, refurbishment and life extension may retain more value than immediate recycling. For PV, glass and aluminium are more readily recovered than high-purity silicon and silver; economics depend on transport, technology and recovered-material quality.

A legal duty to decommission does not guarantee cash will be available decades later if a developer becomes insolvent. Project approvals should require independently reviewed financial assurance: ring-fenced reserves, bonds, insurance or enforceable guarantees, recalculated periodically and protected against insolvency.

Report separately: (1) generation LCOE; (2) adjusted LCOE including major replacements and lifecycle O&M; (3) full project lifecycle cost including decommissioning, recycling and restoration; and (4) system cost including balancing, storage, curtailment and transmission. Apply equivalent lifecycle discipline to nuclear and other firm resources.

5. Existing Government Measures and Their Limits

Existing instruments include the British Industrial Competitiveness Scheme (BICS), British Industry Supercharger, Contracts for Difference, Sizewell C, the SMR programme and Great British Energy. These are relevant building blocks. The remaining question is whether they operate as one coherent mechanism connecting industrial demand, long-term procurement, firm generation, network investment, flexibility and the public balance sheet.

CfDs stabilise generator revenues but do not alone guarantee a competitive delivered price for a particular industrial site. Great British Energy’s current remit is primarily domestic; overseas investment should not be assumed without explicit legal authority and mandate.

6. Proposed UK Industrial Energy Competitiveness Framework

6.1 Long-term industrial power contracts — reduce price and investment uncertainty

Enable 10–20-year contracts for strategic sites, specifying price formula, delivery profile, indexation, balancing, volume flexibility, curtailment, collateral, counterparty obligations and termination rights. Supply may combine nuclear, renewables and imports. Any public support must be transparent, targeted and justified against a published counterfactual.

6.2 Firm low-carbon supply — reduce gas exposure and outage vulnerability

Prioritise safe and economic nuclear life-extension, disciplined delivery of Sizewell C, evidence-led SMR assessment, and domestic engineering and maintenance capability. Complement firm generation with reserves, storage, interconnection and demand flexibility.

6.3 Renewables on full system value — prevent hidden costs

Continue renewable investment where projects demonstrate value after connection, transmission, curtailment, balancing, storage, firm-capacity contribution, land use, lifecycle O&M and decommissioning. Disclose direct footprint and wider site envelope separately.

6.4 International energy and infrastructure portfolio — diversify exposure

Assess overseas upstream gas, LNG, storage, pipelines, generation, transmission and interconnectors, as well as critical equipment and services. Equity ownership or dividends do not guarantee cheaper physical supply. Physical security requires enforceable offtake, transport, terminal, storage and interconnector rights.

Only distributable net cash returns after operating costs, debt service, reinvestment, taxes, reserves and contractual obligations should be considered for transfer to a stabilisation fund. Define legal owner, investment manager, mandate, risk limits, audit, oversight and withdrawal rules before launch.

6.5 Industrial Energy Stabilisation Fund — protect viable capacity during shocks

Limit permitted uses to support for qualifying long-term contracts, temporary rules-based relief during extreme price events, and co-investment in site infrastructure such as direct connections, storage and demand response. Ring-fence, independently audit and disclose fiscal risks. Include eligibility criteria, caps, sunset clauses, liquidity reserves and clawbacks.

6.6 Industrial energy zones — align infrastructure with industrial demand

Assess clusters such as South Wales, Teesside/North East, Humber and North West using industrial load, network constraints, connection queues, logistics, generation and storage proximity, land, skills and credible investment plans. These are candidates, not predetermined winners.

7. Political and International Conditions for Implementation

The programme requires coordination among institutions with different mandates. This table is a coordination map, not a claim that every participant already has all required powers.

Participant	Primary interest	Role in programme
Government / Cabinet	Growth, jobs, energy security	Political leadership; cross-government mandate; resolve institutional conflicts.
HM Treasury	Fiscal sustainability	Control public obligations, guarantees, fund rules and fiscal risk.
DESNZ	Energy policy and decarbonisation	Design policy and coordinate energy and industrial strategy.
Ofgem	Regulation and consumer protection	Independent oversight of regulated market mechanisms within statutory remit.
NESO	System reliability and balancing	Technical assessment of adequacy, constraints, flexibility and network needs.
Great British Energy	Public energy investment	Investment delivery within legal mandate; no assumed overseas powers.
Industrial enterprises	Competitiveness and reliable supply	Define demand, provide data, contract and invest in site flexibility.
Generation companies	Bankable returns	Develop, finance, build, operate and maintain capacity.
Network operators	Infrastructure and regulated returns	Deliver connection, transmission and distribution capacity.
European partners	Trade and energy security	Cross-border trade, interconnector coordination and joint projects.

A programme board must not blur independent regulatory functions. DESNZ and Treasury set policy and fiscal boundaries; NESO supplies technical system analysis; Ofgem retains independent regulatory responsibilities; investment vehicles act under defined mandates; industry supplies demand evidence.

8. Economic Rationale and Evaluation

The programme should be judged by full delivered energy cost and preservation or expansion of productive capacity, not by public money disbursed. Mass subsidies that merely transfer costs to taxpayers or other consumers do not reduce system costs.

- Delivered industrial electricity price, disaggregated into energy, network, balancing, tax and levy components.

- Volatility and hedge effectiveness over rolling 12-, 36- and 60-month periods.
- Price gap against comparable European peers using the same consumption band and tax basis.
- Firm capacity, gas-price sensitivity, reliability, outages and curtailment.
- Connection cost and time, network constraint costs and storage utilisation.
- Industrial output, capacity utilisation, investment, employment and supplier retention.
- Fiscal cost per unit of industrial output preserved or private investment mobilised.
- Portfolio cash yield, liquidity, debt-service coverage, concentration and realised returns.
- Lifecycle O&M, component replacement, decommissioning provision and assurance coverage.

9. Structural Risk: Cheap Automation + Competitive Energy

The cost of industrial robots, automation and AI-enabled process control is falling. Producers combining these technologies with affordable, dependable electricity can reduce unit costs, automate more process stages and improve throughput, quality and energy productivity. The advantage compounds: cheaper automation lowers production costs, while competitive energy makes electrified heat, material handling and other energy-intensive processes more viable.

Lower unit costs allow foreign producers to reduce prices, expand market share and displace British products in domestic and export markets. If the divergence persists, British manufacturers may lose orders, investment, scale, skilled labour and supplier networks. Once those capabilities disappear, rebuilding them may be prohibitively expensive. Failure to respond risks the progressive hollowing-out of British manufacturing.

This is not a deterministic forecast that all British industry will disappear. The risk depends on the persistence of the energy-cost gap, automation adoption, trade exposure, productivity and policy response. It is nevertheless a strategic industrial risk, not a distant technology trend.

10. Stress Testing and Risk Governance

Stress testing is a mandatory design requirement. It should be completed before the overseas portfolio is scaled, before the Stabilisation Fund assumes material obligations and before long-term contracts are offered at scale. Tests should be independently reviewed and repeated annually and after material changes.

Scenario	Illustrative shock	What to test
Gas-price shock	Gas prices double and remain high for 12 months.	Contract affordability, collateral, liquidity, gas-linked exposure.
Low renewable output	Prolonged low wind and solar during cold-weather demand.	Firm capacity, storage duration, imports, demand response, continuity.
Major asset outage	Large generator or critical transmission asset unavailable.	Reserve margins, replacement cost, contract performance, curtailment.
Infrastructure delay	Key network or generation project delayed 3–5 years.	Connection queues, constraints, interim supply, stranded commitments.
Overseas disruption	Asset shut down, expropriated, sanctioned or denied export access.	Offtake rights, jurisdictional concentration, counterparty risk, recovery.
Financial stress	Commodity prices fall, sterling weakens, financing costs rise.	Liquidity, debt service, currency mismatch, support obligations.
Compound crisis	Gas shock, low renewables, interconnector disruption and unit outage.	System adequacy, industrial exposure, emergency prioritisation, fiscal backstop.
Industrial downturn	Recession reduces industrial demand and portfolio income.	Minimum-take contracts, stranded assets, revenue concentration.

These are scenario tests, not forecasts or probability estimates. Shock magnitudes should be calibrated with historical events, market data and independent expert judgement.

10.1 Reverse stress tests

Identify combinations of events that make the framework fail: the minimum cash yield or liquidity at which the Fund cannot meet commitments; the maximum sustained wholesale price at which contracts become unaffordable; portfolio impairment that breaches concentration or solvency limits; and the duration of supply interruption that forces industrial curtailment.

10.2 Minimum safeguards

- No overseas investment without legal due diligence, independent valuation, political-risk assessment, sanctions screening, enforceable offtake/transport analysis and an exit route.
- No claim of physical supply security without contractually secured and operationally tested supply, transport, terminal, storage or interconnector rights.
- Defined liquidity reserves and coverage ratios against committed support obligations.

- Explicit limits by country, counterparty, asset type, currency and commodity.
- Stress-tested collateral, default, indexation, force-majeure and curtailment provisions in long-term contracts.
- Lifecycle cost coverage and independently reviewed end-of-life financial assurance for supported assets.
- Annual public risk report with scenario results, sensitivities, exposures, realised returns and corrective actions.

Proceed through stage gates: design and data validation; limited pilot; independent evaluation; scale-up only if tests pass. A failed stress test should trigger redesign or a stop decision, not automatic taxpayer absorption of losses.

11. Implementation Path and Pilot

A 24–36-month pilot in one industrial region should focus on data, contracts, governance and limited infrastructure measures—not assume that major generation or transmission projects can be completed within the period.

- Select a region using industrial load, network constraints, connection queues, land availability and committed participants.
- Establish a confidential baseline: hourly consumption, delivered prices, contracts, outages, flexibility and planned investment.
- Design and test a small portfolio of long-term contracts with different supply and indexation structures.
- Implement one or two measurable site-level measures, such as a direct connection, storage or demand response.
- Complete legal, financial and physical-security due diligence for candidate overseas assets; simulate and stress-test before material capital commitment.
- Introduce lifecycle cost disclosure and financial-assurance requirements for supported assets.
- Evaluate independently against a published counterfactual, including fiscal and distributional effects.

Success should be measured by delivered price versus counterfactual, volatility, reliability, connection cost and time, private investment, industrial output and fiscal cost per unit of output preserved. A pilot fails if apparent savings merely transfer unpriced risks to taxpayers or other consumers.

12. Principal Risks

Principal risks include prolonged high gas prices; nuclear construction delays and cost overruns; correlated low-renewable output; transmission delays; land-use conflict; incomplete lifecycle estimates; inadequate decommissioning security; fiscal constraints; overseas political and regulatory risk; currency and commodity exposure; delayed or impaired portfolio distributions; counterparty failure; and Fund obligations exceeding liquid resources.

Diversification and a new fund do not eliminate these risks. They must be quantified, allocated to parties capable of managing them, reflected in contract prices and subjected to explicit limits. The framework must retain the ability to decline an investment or stop expansion.

13. Conclusion

High industrial electricity costs are a long-term threat to the productive base of the United Kingdom. Higher energy costs raise unit production costs; higher costs weaken price competitiveness; persistent loss of competitiveness can displace British output from domestic and international markets.

A coherent framework should connect industrial procurement, firm supply, renewables, networks, land-use planning, lifecycle liabilities, international energy exposure and public investment risk. It is not a case for abandoning renewables or relying on nuclear alone. Every option should be assessed by its contribution to reliable, affordable electricity at the industrial site, including system, spatial, lifecycle and fiscal costs.

The international portfolio proposal is conditional: ownership alone does not guarantee physical supply, and returns may not arrive when industry needs support. Contractual rights, legal authority, portfolio limits and rigorous stress tests are prerequisites.

The practical first step is a regional pilot under coordinated oversight by DESNZ, HM Treasury, relevant investment institutions, NESO and Ofgem, with active industrial participation. Scaling should follow only after independent evaluation shows that the framework reduces the full cost and risk of industrial energy rather than merely redistributing them.

Principal Sources and Methodology

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Methodological notes

European comparison: 20–70 GWh/year industrial consumers, H2 2025. Values converted from p/kWh using 2025 annual-average GBP/USD of approximately 1.34. Conversion is for unit comparison, not local purchasing-power equivalence.

Global benchmark: selected IEA regional estimates for large energy-intensive consumers, generally above 150 GWh/year, 2024. Percentages against UK are arithmetic context because cohort, period and regional basis differ.

Policy costs: the industrial 4.5 p/kWh levy increase is compared with net 5.4 p/kWh increase across energy/supply, network and policy-cost components. It is not described as percentage growth in levies. The IFS 59% figure is for households and is not applied to industry.

Renewables: module efficiency is not capacity factor, annual yield or firm-capacity contribution. Land estimates distinguish direct footprint from wider project envelope.

Lifecycle estimates are indicative and depend on design, geography, maintenance regime and scope. Stress scenarios are illustrative tests, not probability-weighted forecasts.